

SANJAY P

AI/ML Engineer | Python | Machine Learning | Deep Learning | NLP

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Professional Summary

I'm an AI/ML enthusiast with a strong foundation in machine learning, deep learning, and natural language processing, gained through hands-on experience with Python, TensorFlow, PyTorch, and Hugging Face Transformers. I've explored fine-tuning large language models (LLMs), building Retrieval-Augmented Generation (RAG) systems, and deploying AI solutions using inference APIs.

My projects—including intelligent chatbots, predictive applications, and document-aware assistants—reflect my drive to apply AI in solving real-world problems. I'm passionate about continuous learning and advancing my expertise in generative AI and mathematical reasoning models. Aspiring to launch my career in AI engineering or data science, I'm eager to collaborate, innovate, and contribute to cutting-edge AI initiatives that create meaningful impact.

Technical Skills

- **Languages:** Python, Java, JavaScript (ES6+), SQL
- **AI/ML Frameworks:** TensorFlow, PyTorch, Scikit-learn, Hugging Face Transformers (fine-tuning, Inference API)
- **NLP & Generative AI:** NLTK, LLMs (GPT, BERT, gpt-oss-20b), LangChain, RASA
- **Data Tools:** Pandas, NumPy, Matplotlib, Power BI
- **Web Frameworks:** Django, Flask, React, Next.js, Spring Boot
- **Databases:** MySQL, MongoDB
- **Tools:** Git, Docker, AWS (basic), CI/CD

Work Experience

Junior Associate Program Manager - Iamneo Coimbatore, Tamil Nadu May 2024 – Present

- Led cross-functional support operations using Jira and Zoho Desk, achieving 95%+ SLA compliance while resolving AI-related client queries efficiently.
- Utilized Power BI and Excel for data analytics, creating AI-driven dashboards to enhance product features like personalized learning.

- Spearheaded end-to-end support for AI-integrated web platforms, contributing to sprint planning, QA testing of ML models, and UX improvements for chat interfaces.
- Organized NERD programs and codeathons focused on AI, optimizing algorithms to reduce response times by 20%.

Software Engineering Intern - Virtusa Remote Jan 2023 – Mar 2023

- Gained practical exposure to full-stack development with AI focus using Angular, Node.js, Java, Spring Boot, and MongoDB.
- Developed and deployed mini-projects in Agile sprints, incorporating ML preprocessing with Pandas and version control via Git/CI/CD.
- Built proof-of-concept AI applications, including sentiment analysis using NLTK and Hugging Face Transformers.

Key Projects

Math Solver (Ongoing - AI Reasoning)

- Developing a math solver using Aryabhata-1.0 (7B model) for JEE-level problems, fine-tuned with Hugging Face Transformers.
- Deploying via Inference API for step-by-step solutions, achieving 85% accuracy on benchmark datasets.
- GitHub: [In Progress Repository Link]

RAG-Based Document Chatbot (NLP/Generative AI)

- Built a chatbot using gpt-oss-20b for document analysis with Retrieval-Augmented Generation (RAG) via LangChain.
- Fine-tuned with Hugging Face, reducing hallucination by 30% on test datasets; deployed with Docker.
- GitHub: [Link to Repository]

Weather App (Predictive Analytics)

- Developed a responsive Next.js/Tailwind app using OpenWeather API, enhanced with PyTorch for weather forecasting (low MAE).
- Visualized trends with Plotly; deployed on Vercel.
- GitHub: <https://github.com/CopyCoderJay/Weatherapp>

Plant Identification App (Computer Vision)

- Full-stack app with React frontend, Django backend, and TensorFlow.js CNN (92% accuracy on 10k+ images).
- Integrated Hugging Face Inference API for real-time predictions; deployed on AWS.
- GitHub: [Link]

Covid Chatbot (Conversational AI)

- Created a RASA-based chatbot for real-time COVID-19 information, using Hugging Face Transformers for intent recognition (85% accuracy).
- Integrated generative responses with OpenAI API; deployed with Docker.
- GitHub: <https://github.com/sanjay-dixit07/CHATBOT-RASA>

Certifications & Achievements

- Top 30 Finalist - IIT Kharagpur Data Science Hackathon (2021): PyTorch-based anomaly detection.
- Finalist - Project Presentation, Goa Institute of Management (2021): NLP sentiment analysis tool.
- Runner-up - ML Hackathon, Techgig & Times Internet (2021): Scikit-learn/ TensorFlow stock prediction.
- Kaggle: Top 20% in Titanic Survival; Improved NLP Disaster Tweets models with Hugging Face.
- Certifications: Coursera Deep Learning Specialization; Hugging Face NLP Course.

Education

B.Tech - Computer Science & Business Systems

Bannari Amman Institute of Technology, Erode

2019 – 2023

CGPA: 8.43 / 10

- Key Courses: Artificial Intelligence, Machine Learning, Business Management, Algorithms
- Capstone Project: Resume Builder

Languages

- English: Full Professional Proficiency
- Tamil: Native/Bilingual Proficiency